A factory produces two types of products, P1 and P2. Each product requires processing time on two machines: Machine A and Machine B. The processing times (in hours per unit) and the profits (in dollars per unit) are given below:

Product	Machine A (hours)	Machine B (hours)	Profit (dollars)
P_1	2	3	40
P_2	3	2	50

The factory has the following constraints:

- Machine A is available for a maximum of 120 hours.
- Machine B is available for a maximum of 90 hours.

The objective is to maximize profit.

Based on the previous solution, answer the following questions:

Resource Availability:

- How would the solution change if Machine A's availability increased to 150 hours?
- What happens if Machine B's availability decreases to 80 hours?

Profit Changes:

- If the profit for P2 increases to $60 per unit, does the optimal solution change?
- If the profit for P1 decreases to $30 per unit, what is the new optimal solution?

Constraint Impact:

- What happens to the feasible region if a new constraint $x1+x2 \leq 50$ is added?
- If one constraint is removed (e.g., Machine A's constraint), how does it affect the problem?

Last, make a spider plot considering if the machine A have 130, 140, 150 and 160 hours and the machine B have 100, 110, 120 and 130 hours. (Suppose that the other machine hours remain the same during your scenario).